



Engineering Technical Assessment

1. Welcome

Hello and welcome to the Eye Technical Assessment!

Firstly, thank you for taking the time to do this technical assessment. It is an important part of recruitment process, as you can imagine. If you have got this far in our interview process, then you have impressed us and we would now like to give you the opportunity to further impress us with your technical skillset.... that is a big reason of why we are at this point in any case.

2. Why this kind of Technical Assessment

We have chosen to proceed with a take-home assessment. This will be followed up by an interview, performed by 2 senior engineers who will talk through your code with you (we are trying to replicate the Code Review process as far as possible to simulate what it would be like to work with you). We have put in a lot of thought surrounding what kind of tech assessment that we would like to make use of before settling on this as an option. The primary reason is that we are looking to both check your skillset and importantly to simulate what it will be like to work with you. Both parts are equally important to us.

In doing this, we invite you to do the same with us. In this process, you should hopefully also get a good sense of two of your potential-future-colleagues and get a glimpse of what it would be like to work as an engineer at Eye Security. Please take some time to think of if you like what you see and can envisage yourself as a part of the team.

PS. We are all about cyber security and wanted to represent that in the tech assessment.

3. Rules

1. We would like you to timebox this assessment to **no more than 4 hours**.
2. Please use **one** of the following languages:
 - Go
 - TypeScript
 - Java/Kotlin
 - C#
 - Python
3. We are not looking for a fully completed solution. Instead, we value a well thought, high-quality implementation that demonstrates your engineering approach and decision-making process in the given time frame. The goal of this assessment is to understand how you think and structure your work. A well-designed, well-structured and thoughtful partial solution is far more valuable than a rushed, fully completed one.
4. The use of LLM / AI-assistance in the assessment is completely fine. Just be aware that you will be expected to own any code produced and talk through any code as if it were your own.
5. We are not expecting you to deploy the code or implement any cloud infrastructure - if implementation details are necessary for your answer, then you can simply state the implementation details in text in the code (eg. they can be added to the README.md).
6. Where necessary, please state your assumptions in the code.
7. Authentication/authorization of your API are out of scope
8. Please commit the code to a repository in Github / GitLab and send us a link where we can clone it at least 24 hours before the interview.
9. In the interview connected with this tech assessment expect to answer questions on your code, certain design and architectural choices, etc.

4. Jira Ticket

The Cyber Security Operations team wants to populate the Analytics Service by uploading malicious activity logs (in csv format). This will be their primary way of ingesting data into the analytical systems used for Incident Response. This makes it a mission-critical system.

In the refinement (when you were away), we decided that the best way to accomplish this is to:

1. Build a CLI application that will ingest the csv file.
 - a. [Nice to Have] Allow CLI to include a parameter to filter the records that are to be ingested.
 - b. Provide feedback to the CLI user.
2. Build a Microservice API project to process the records. It must have a single endpoint that will receive the call from the CLI application.
 - a. For each record, the data needs to be enriched by calling the Enrichment Service ([docs](#))
 - i. Heads up - we have had intermittent issues with the Enrichment Service before. We are unsure if Canary Team has resolved their issue yet. Proceed with caution.
 - b. The records must be sent to the Analytics Service ([docs](#)).
 - i. After the budget incident, this service is now rate limited at 20 messages per 10 seconds.

We use Authentication for both the Enrichment Service and the Analytics Service. Authentication can be done via this HTTP header: Authorization: eye-am-hiring

SecOps has provided us with this csv file as an example of the data that they are expecting.

[attached document example_data_2.csv]

Please timebox this to 4 hours.

5. Good to go

That is about it. Thank you for spending some of your time in completing this tech assessment. Good luck!